

Fourier analysis – 1

Fourier series: convergence and summation

Homework

13 сентября 2026 года

Theoretical minimum. The partial sum is the convolution with the Dirichlet kernel, so pointwise convergence is a local question (localisation) and is guaranteed by Dini's or Jordan's criterion. Summation by a positive kernel always works: the Fejér and Abel–Poisson means of a continuous function converge to it uniformly, and this gives both the uniqueness theorem and the density of trigonometric polynomials. The Dirichlet kernel is not positive, its L^1 norms grow like $\ln N$, and therefore a continuous function may have a divergent Fourier series at a point — the proof is Banach–Steinhaus, not a construction. Near a jump the partial sums always overshoot by about nine per cent of the jump. In L^2 everything is easy: the partial sums are the orthogonal projections and converge in the mean by Parseval.

1. Compute the Fourier series of the sawtooth $f(x) = \frac{\pi-x}{2}$ on $(0, 2\pi)$, prove that $S_N f(\frac{\pi}{N}) \rightarrow \int_0^\pi \frac{\sin t}{t} dt$ as $N \rightarrow \infty$, and deduce that the partial sums overshoot the jump by the factor $\frac{2}{\pi} \int_0^\pi \frac{\sin t}{t} dt - 1 \approx 0.0895$.
2. Prove Jordan's criterion: for a function of bounded variation the Fourier series converges at every point to the half-sum of the one-sided limits.
3. (Карацёв 1.74) Let $f \in L^1[-\pi, \pi]$ and let $T_n(f, x)$ be the partial sums of its Fourier series. Prove that $\int_a^b T_n(f, x) dx \rightarrow \int_a^b f(x) dx$ for all $a, b \in [-\pi, \pi]$. (The case $f \in L^2$ is easier — do it first.)
4. (Карацёв 1.69) Let $x_1, \dots, x_N$ be distinct points of $(-\pi, \pi]$ and let $a_1, \dots, a_N \in \mathbb{C}$ be not all zero. Prove that the sequence $c_n = a_1 e^{inx_1} + \dots + a_N e^{inx_N}$ cannot tend to zero as $n \rightarrow \infty$.

5. Expand x , x^2 and $|x|$ on $[-\pi, \pi]$ into Fourier series, and deduce the values of $\sum \frac{(-1)^{n-1}}{n}$, $\zeta(2)$ and $\sum \frac{1}{(2k+1)^2}$; say which of these follow from pointwise convergence and which from Parseval.
6. Prove that if the Fourier series of an integrable function converges uniformly, then its sum is that function; and give an example of a trigonometric series which converges everywhere but is not the Fourier series of any integrable function.
7. Prove the sharper form of localisation: if f vanishes on (a, b) , then $S_N f \rightarrow 0$ uniformly on every compact subset of (a, b) , with a rate depending only on the distance to the endpoints.
8. Prove that the Fejér means of a continuous function are uniformly bounded by $\max |f|$ while the partial sums need not be, and construct a continuous function with $\limsup_N \|S_N f\|_\infty = \infty$.
9. Prove that for $f \in L^2$ the partial sums converge to f in the mean square and that $S_N f$ is the best L^2 approximation by trigonometric polynomials of degree N ; then exhibit a continuous function for which the convergence is not uniform.
10. State Kolmogorov's theorem on an L^1 function with everywhere divergent Fourier series and Carleson's theorem on almost everywhere convergence for L^2 , and explain what each of them says about the practical question "may I differentiate or evaluate a Fourier series pointwise".